

SOFTWARE ENGINEERING LAB

Lab 1: Requirements Engineering & UML Use-Case Modelling

PROBLEM STATEMENT #27 | Smart Cities, Transport & Logistics

Warehouse Inventory & Pallet Location Tracker

Functional Requirements (FR-001 to FR-005)

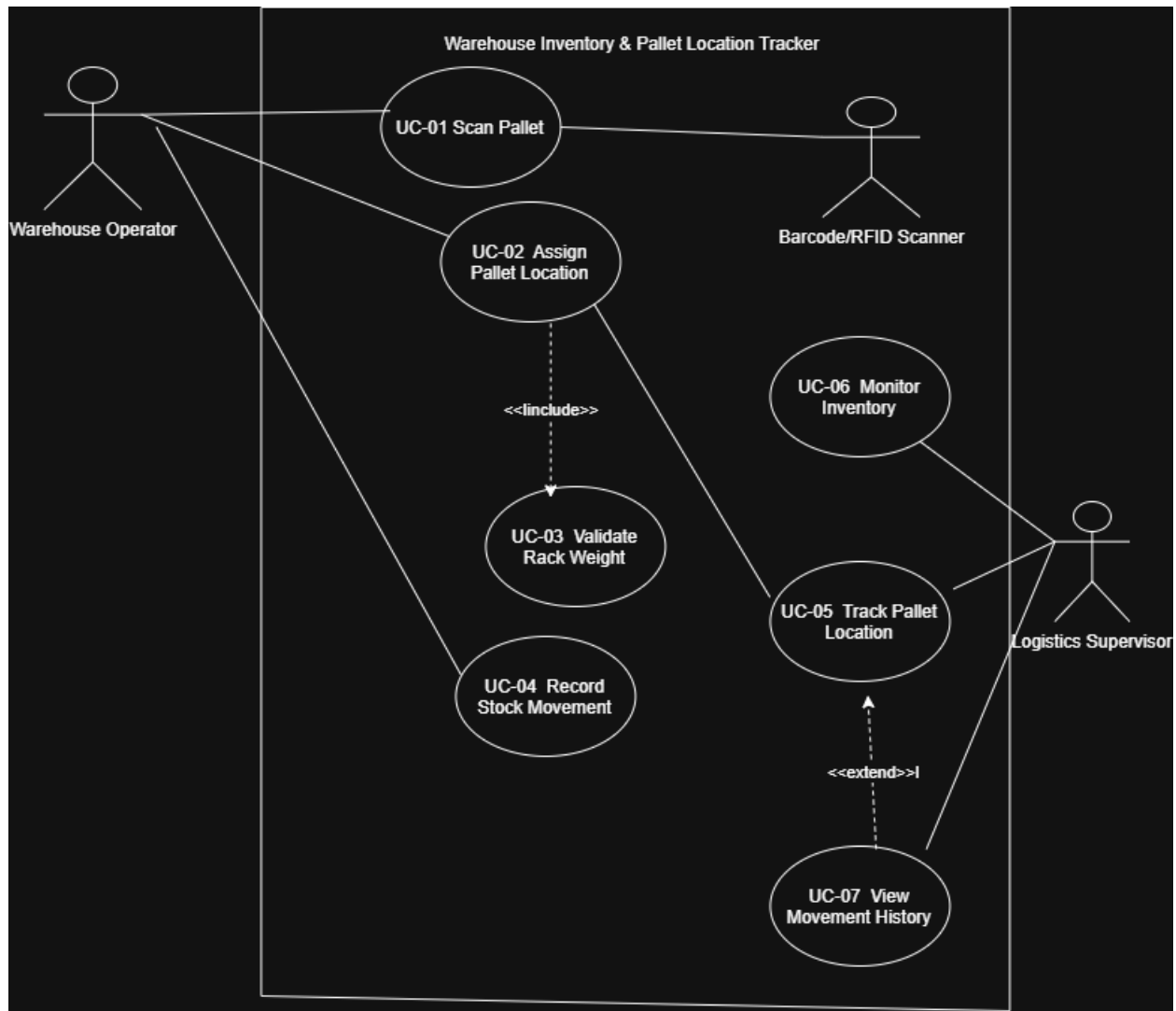
Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall validate proposed pallet placement coordinates against the rack maximum load capacity before confirming storage.	High	Pass: The system confirms pallet placement only when the total rack weight after placement is within the rack's rated maximum capacity. Fail: The system permits placement when the rack's maximum load capacity would be exceeded.	Prevents unsafe pallet placement and ensures rack weight limits are enforced.
FR-002	Functional	The system shall identify a pallet when its barcode or RFID tag is scanned by the warehouse operator.	High	Pass: A valid barcode/RFID scan displays the corresponding pallet ID and inventory details. Fail: An invalid or unreadable identifier is accepted without displaying an error.	Enables accurate pallet identification and reduces manual data entry.
FR-003	Functional	The system shall assign and store a valid three-dimensional shelf location for each pallet after successful storage confirmation.	High	Pass: The system stores the pallet's rack, level, and position coordinates after successful placement. Fail: The system confirms storage without recording the complete 3D location.	Ensures every stored pallet can be accurately located in the warehouse.
FR-004	Functional	The system shall record each pallet movement along with the pallet identifier, previous location, new location, and movement time.	High	Pass: After a pallet is moved, the system records its ID, previous location, new location, and timestamp. Fail: A completed pallet movement is recorded without any required detail.	Maintains an accurate history of stock movements for tracking and auditing.
FR-005	Functional	The system shall allow an authorized user to retrieve the current warehouse location of a pallet or SKU.	High	Pass: Entering a valid pallet ID or SKU returns its current rack, level, and position. Fail: The system returns an incorrect or incomplete location	Helps warehouse staff quickly locate stored inventory.

				for a valid pallet or SKU.	
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2 Non-Functional Requirements (NFR-001 & NFR-002)

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
NFR-001	Nonfunctional	The warehouse item lookup service shall locate any SKU pallet coordinate across 100,000 bins in under 100 ms.	High	Pass: Benchmarking confirms that SKU pallet-location lookup completes in less than 100 ms under the defined test load. Fail: Any tested lookup takes 100 ms or longer.	Ensures fast inventory lookup even with a large number of storage bins.
NFR-002	Nonfunctional	The system shall preserve successfully recorded pallet movement transactions without loss during normal warehouse operation.	High	Pass: In a test of 1,000 successfully completed pallet movements, all 1,000 movement records remain retrievable. Fail: Any successfully recorded movement is missing or unretrievable.	Ensures reliable inventory records and prevents loss of stock movement information.

UML Use-Case Diagram



Use case Flow Specialization

Use Case: UC-02 — Assign Pallet Location

Primary Actor: Warehouse Operator

Supporting Actor: Barcode/RFID Scanner

Preconditions

1. The Warehouse Operator is authorized to use the warehouse inventory system.
2. The pallet has been identified using its barcode or RFID tag.
3. The pallet's identification and weight information are available in the system.
4. A proposed rack location is available for validation.

Postconditions

Success:

1. The pallet is assigned to a valid three-dimensional rack location.
2. The pallet's current location is updated in the inventory system.
3. The successful stock movement is recorded with the required movement details.

Failure:

1. The pallet is not assigned to an invalid or overloaded rack location.
2. The system displays the reason for rejecting the proposed location.
3. The Warehouse Operator can select another location.

Main Success Scenario

1. The Warehouse Operator scans the pallet's barcode/RFID tag.
2. The system identifies the pallet and retrieves its inventory details.
3. The Warehouse Operator selects or enters the proposed rack, level, and position.
4. The system validates that the proposed coordinates represent a valid warehouse location.
5. The system checks the rack's current load and the pallet's weight.
6. The system confirms that placing the pallet will not exceed the rack's maximum load capacity.
7. The system confirms the pallet placement.
8. The system stores the pallet's rack, level, and position as its current location.
9. The system records the stock movement with the pallet ID, previous location, new location, and timestamp.
10. The system displays a successful placement confirmation.

Alternate Flow — Rack Weight Limit Exceeded

A1. Rack exceeds maximum permitted load

- At Step 6, the system determines that placing the pallet would exceed the rack's maximum load capacity.
- The system rejects the proposed pallet placement.
- The system displays a message indicating that the rack cannot safely accommodate the pallet.
- The Warehouse Operator selects another proposed rack location.

- The system validates the new location and its rack capacity.
- If the new location is valid, the system continues from Step 7 of the Main Success Scenario.